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United States Patent	12415038
Kind Code	B2
Date of Patent	September 16, 2025
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Endoscopic cannula

Abstract

A cannula comprising a hollow tube made of a biocompatible, thermoplastic polymer, memory shape material, tipped with an injection needle or an electrode at it distal end, assumes desired shapes as it distally emerges from a lumen or channel of an instrument such as an introducer or endoscope. The desires shapes include a curvature conforming to an inverse tangent function. The cannula may be supplied alone, with or without an injection needle or electrode, and in a sterile pouch, for use with a device of a user's choice that has a suitable lumen or channel to guide the cannula, of the cannula may be used without such an instrument. The injection needle can be shorter than conventional to thereby inject medication only or primarily in submucosa and can be sharpened only or mostly at it distal end while the remaining portion of a bevel is not sharp, to thereby reduce the size of an opening the needle makes in tissue and helps keep the medication from leaking out of the injection site.

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Appl. No.: **17/208955**

Filed: **March 22, 2021**

Prior Publication Data

Document Identifier	Publication Date
US 20210299358 A1	Sep. 30, 2021

Related U.S. Application Data

continuation parent-doc US 15834333 20171207 US 10463797 child-doc US PCT/US2018/050087
continuation-in-part parent-doc US 16564668 20190909 US 10953159 child-doc US 17208955
continuation-in-part parent-doc WO PCT/US2018/050087 20180907 PENDING child-doc US 16564668
continuation-in-part parent-doc US 15834333 20171207 US 10463797 child-doc US 16564668
continuation-in-part parent-doc US 15722168 20171002 US 10456164 child-doc US 16564668
continuation-in-part parent-doc US 15697640 20170907 US 10286459 child-doc US 15834333

Publication Classification

Int. Cl.: **A61B1/307** (20060101); **A61B1/00** (20060101); **A61M5/31** (20060101); **A61M5/315** (20060101)

U.S. Cl.:

CPC **A61M5/3137** (20130101); **A61B1/00135** (20130101); **A61B1/307** (20130101); **A61M5/3156** (20130101); **A61M5/31595** (20130101); A61M2005/3131 (20130101); A61M2205/3379 (20130101); A61M2205/581 (20130101); A61M2205/582 (20130101); A61M2210/1085 (20130101)

Field of Classification Search

CPC: A61B (1/307); A61B (1/00135); A61B (1/018); A61M (2210/1085); A61M (5/3137); A61M (5/3156); A61M (2005/3131); A61M (5/31595); A61M (2205/3379); A61M (2205/581); A61M (2205/582)

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Background/Summary

REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. patent application Ser. No. 16/564,668 filed Sep. 9, 2019 and scheduled to issue as U.S. Pat. No. 10,953,159 on Mar. 23, 2021, which is a continuation-in-part of each of (a) U.S. patent application Ser. No. 15/722,168 filed Oct. 2, 2017, (b) U.S. patent application Ser. No. 15/834,333 filed Dec. 7, 2017, and (c) International Patent Appl. No. PCT/US18/50087 filed Sep. 7, 2018 published as WO 2019/051315 on Mar. 14, 2019. Said International Patent Appl. claims priority to said U.S. patent application Ser. No. 15/834,333 as well as to U.S. patent application Ser. No. 15/722,168 filed Oct. 2, 2017, and said U.S. patent application Ser. No. 15/834,333 is a continuation-in-part application of U.S. patent application Ser. No. 15/697,640 filed Sep. 7, 2017 and issued as U.S. Pat. No. 10,286,159 on May 14, 2019. (2) This application incorporates by reference and claims the benefit of the filing date of each of the above-identified patent applications, as well as of the applications that they incorporate by reference, directly or indirectly, and the benefit of which they claim.

FIELD
(1) This patent specification pertains to cannulas used primarily in medical instruments and procedures although they may have non-medical uses as well.
BACKGROUND
(2) Flexible cannulas are widely used in medical procedures to pass through cannula lumens in instruments such as endoscopes and introducers and treat internal patient sites, for example to inject a substance in an internal organ, to deliver a substance to an internal site, to apply electrical current or other energy to an internal site, to view an internal organ, and to extract tissue samples or fluids from an internal site, to name a few. Such cannulas may also find use in fields other than medicine, for example to access internal regions in objects that are not otherwise easily reachable.
(3) One condition treated with injections in the bladder Overactive Bladder (OAB). In about 2013, the FDA approved treatment of OAB by periodically injecting OnabotulinumtoxinA (Botox) at sites in a pattern along the inner wall of the patient's bladder. Previous methods of delivering OnabotulinumtoxinA to the bladder have involved inserting a cystoscope in the bladder through the urethra, passing a long metal needle through a lumen or working channel in the cystoscope, and manipulating the entire assembly both laterally and along the axis of the urethra as a unit to inject medication at numerous injection sites in the bladder. Such previous devices and methods have resulted in significant patient discomfort and hold the danger of damage to the urethra and surrounding tissue.
(4) A desired placement and pattern of the multiple injections in the bladder are associated with significantly improved treatment outcomes. It is important that devices and methods of injecting OnabotulinumtoxinA into the bladder offer physicians performing the procedure precise control. However, it had been difficult to create precise injection patterns using such previous devices and methods when the scope moves with the needle when aiming for a new injection site. Moreover, said devices are usually not disposable, and must be disassembled and sterilized after each use, making them difficult to maintain and increasing the risk of contamination or infection. More recently, disposable cannulas with integrated video camera and injection needles have been discussed. See for example US Published Application 2016/0367119 A1 and U.S. Pat. No. 10,278,563.
(5) There are proposals to use cannulas made of shape memory metals that change shape after emerging from lumens in devices such as introducers. See Sachveda U.S. Pat. No. 5,607,435 illustrating in FIG. **4** a tubular section **42** with a leading edge shaped as a surgical needle that curls after exiting a straight delivery tube **50**. According to the patent, the curling section is made of metal alloys such as Nitinol. See also McGucken Published Patent Application US 2003/0032929 A1 proposing a shape memory Nitinol infusion needle and mentioning that polymer material might be used in some applications for a shaft in which a shape memory stylet moves but not that the

polymer material can be shape memory.

SUMMARY OF THE DISCLOSURE

(6) According to some embodiments, a medical device for insertion into a patient's body comprises: an elongated hollow cannula made of a biocompatible, non-metallic material; said cannula having a first shape when confined by first external forces acting thereon; said cannula having a distal portion that reverts to a second shape when free of said first external forces, and said distal portion having a distal end; said second shape having a predetermined curvature defined by an inverse tangent function and comprising two bends in different directions; whereby said distal end of the cannula is radially spaced from an axis of said first shape when said distal portion of the cannula has reverted to said second shape; and wherein said cannula is sufficiently flexible to assume said first shape when in a working channel of an introducer or endoscope but at least said distal portion of the cannula has a flexural modulus that is sufficiently high when in said second shape to force said distal end against tissue for selected medical action on said tissue when a selected distal force is exerted on a proximal portion of the cannula of the introducer or endoscope. The device can further include an injection needle protruding from said distal end of the cannula, wherein and said distal portion of the cannula has a sufficiently high flexural modulus for inserting said needle into patient tissue after the distal portion of the cannula has reverted to said second shape. The needle has a beveled distal portion with a distal end that is sharpened to make a hole in tissue and a proximal portion that is sufficiently dull to dilate rather than cut tissue at the hole as said proximal portion of the bevel is inserted in said hole. The needle can protrude distally a distal end of the cannula by roughly 3 mm and said distal end of the cannula surrounds the needle and acts as a stop limiting the needle penetration depth in said tissue. The cannula and injection needle can be configured to limit depth of penetration into tissue to inject medication in submucosa rather than underlying muscle. The injection needle can be shaped and configured to form a hole in tissue by a distal tip of the needle and to dilate rather than cut tissue around the hole as a portion of an injection needle portion that is proximal from said distal tip penetrates the issue. The device can comprise an electrode rather than a needle protruding distally from the cannula and an electrical connection from the electrode to a proximal portion of the cannula, or another surgical device protruding distally from said cannula. The device can comprise a guide wire in said cannula configured for motion relative to the cannula along a length of the cannula. The distal portion of said cannula has a flexural modulus of about 595,000 psi, and the entire cannula can comprise a single hollow tube made of PEEK. The two bends of the distal portion of said cannula can be spaced from each other by a straight portion of the cannula, and the two bends can be in opposite directions.

(7) According to some embodiments, a method comprises: inserting an elongated hollow cannula made of a biocompatible, non-metallic material into a passage in a patient's body while confining at least a distal portion of said cannula to a first shape by first external forces acting thereon; thereafter gradually moving a distal end of said cannula distally from said confining to thereby cause said distal portion of said cannula to gradually revert to a second shape that has a predetermined curvature defined by an inverse tangent function and comprises two bends in different directions and cause said distal end to move radially from an axis of said first shape; and applying medical action at selected sites of internal tissue of said patient with device protruding distally from a distal end of said cannula; wherein said injection sites are spaced from each other by distances that depend on a distal distance of said device from said first external forces. The cannula can be confined by enclosing it in a working channel of an introducer or endoscope. The medical action can comprise injecting medication at said sites or applying electromagnetic or ultrasound energy or heat to patient tissue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a side view of a medical injection assembly, according to some embodiments.
(2) FIG. **2** is a cross-section of a sheath or introducer, according to some embodiments.
(3) FIG. **3** is a cut-away isometric view of a sheath or introducer, according to some embodiments.
(4) FIG. **3***a* illustrates a distal portion of a biocompatible polymer cannula that has reverted to a curvature conforming to an inverse tangent function after exiting a distal end of an introducer, according to some embodiments.
(5) FIG. **3***b* illustrates a distal portion of a biocompatible polymer cannula that has reverted to a curvature of over 90 degrees after exiting a distal end of an introducer, according to some embodiments.
(6) FIG. **3***c* illustrates a distal portion of a biocompatible polymer cannula that has a curvature conforming to an inverse tangent function and can be used with alternative introducers or endoscopes that have a cannula lumen or a working channel, or by itself, with or without an injection needle at its distal tip, according to some embodiments.
(7) FIG. **3***d* illustrates a distal portion of a biocompatible polymer cannula that has a curvature of over 90 degrees and can be used with alternative introducers or endoscopes that have a cannula lumen or a working channel, or by itself, with or without an injection needle at its distal tip, according to some embodiments.
(8) FIG. **3***e* illustrates a distal portion of a biocompatible polymer cannula that has a curvature of less than 90 degrees and can be used with alternative introducers or endoscopes that have a cannula lumen or a working channel, or by itself, with or without an injection needle at its distal tip, according to some embodiments.
(9) FIG. **4** is a cross-section of a handle of an introducer, according to some embodiments.
(10) FIG. **5**A is a side view of a cannula exiting a sheath or introducer, according to some embodiments.
(11) FIG. **5**B is a side view of a cannula exiting a sheath or introducer, according to some embodiments.
(12) FIG. **5**C is a side view of a cannula exiting a sheath or introducer, according to some embodiments.
(13) FIG. **6** is an isometric view of a plunger body, according to some embodiments.
(14) FIG. **7** is a bird's eye view of a finger grip, according to some embodiments.
(15) FIGS. **7***a* and **7***b* are two perspective views from different viewpoints that illustrate another example of a syringe that can be coupled to a fluid connector.
(16) FIG. **8**A is a frontal view diagram of a bladder injection pattern, according to some embodiments.
(17) FIG. **8**B is a side view diagram of a bladder injection pattern, according to some embodiments.
(18) FIG. **9** is a perspective view of an injection needle with a special bevel at its distal end, according to some embodiments.
(19) FIG. **10** is a sectional view of an injection needle according to some embodiments.
(20) FIG. **11** is like FIG. **3**A but shows a medical device other than an injection needle protruding distally from a cannula, according to some embodiments.
DETAILED DESCRIPTION
(21) A detailed description of examples of preferred embodiments is provided below. While several embodiments are described, the new subject matter described in this patent specification is not limited to any one embodiment or combination of embodiments described herein, but instead encompasses numerous alternatives, modifications, and equivalents. In addition, while numerous specific details are set forth in the following description in order to provide a thorough understanding, some embodiments can be practiced without some or all these details. Moreover, for the purpose of clarity, certain technical material that is known in the related art has not been

described in detail in order to avoid unnecessarily obscuring the new subject matter described herein. Individual features of one or several of the specific embodiments described herein can be used in combination with features of other described embodiments or with other features. Further, like reference numbers and designations in the various drawings indicate like elements.

(22) This patent specification describes special, hollow, biocompatible polymer cannulas configured to pass through lumens or working channels of a variety of devices and revert to desired predefined curvatures after distally exiting. The lumens of channels can be straight or curved differently from the predefined curvatures to which the polymer cannulas revert when out of those lumens or channels. The lumens or channels can be in any one of a great variety of instruments, including without limitation cystoscopes and other endoscopes of introducers. The curvatures to which the cannulas revert can be as suitable for a specific need or application. The curvature to which a cannula reverts can have a single bend of any desired angle up to 180 degrees relative to a long axis of the cannula, and in some cases can exceed 180 degrees. Another curvature to which a cannula reverts can have two bends in opposite direction, i.e., the curvature can conform to an inverse tangent function. Other curvatures also are possible, e.g., more than two bends, or bends in different planes, etc. The cannulas typically have a distal portion that reverts to the predefined curvatures while the rest of the cannulas are straight or have different curvatures. The cannulas typically have proximal ends that are accessible, for example to inject fluids or other substances into the proximal end of the cannulas or to withdraw substances through the proximal ends. For example, the proximal ends of the cannula can extend proximally from the lumens or working channels for such access.

(23) For use of such cannulas to inject substances at an internal site in a patient, injection needles can be secured to distal tips of the cannulas. For other purposes, such as infusing a substance into or withdrawing a substance from an internal site, the distal tips of the cannulas may be simply open or may be shaped as desired or needed. A cannula that is open at its distal end can be used to pass a guide wire therethrough, and once the guide wire is in place the cannula can be withdrawn, leaving the guide wire in the patient such than additional devices or instruments can be passed over the guide wire to a desired anatomical location. The cannulas can be tipped with electrodes that can be flexible or can be Bugbee type. The entire cannula or a portion of a cannula can be opaque to x-rays to be visible in x-ray imaging.

(24) The lumens or working channels through which the cannulas pass can be in instruments such as endoscopes, introducers, and other devices that can guide cannulas and allow them to protrude distally from the device. Such devices can include viewing components, such as a video camera and a light source at a distal end, or a scope channel in which a viewing device can be inserted and may include one or more lumens in addition for the lumen or working channel for the cannulas, e.g., fluid passages or working channels for surgical instruments. Alternatively, the devices guiding the cannulas can be simply guides, with no viewing facilities and may or may not have any additional lumens or channels.

(25) One example of such cannulas, as used to inject Botox or other medication into a pattern of injection sides at the inner wall of a patient's bladder, is described in commonly owned U.S. patent application Ser. No. 15/697,640, now U.S. Pat. No. 10,286,159, to which this patent specification claims priority. This patent specification recapitulates that example as illustrative of one use of such cannulas and adds disclosure of other examples of uses of such cannulas. In other examples, the cannulas can be used to access other organs such as the uterus, the ureters, and the kidneys. In still other examples, the cannulas can be tipped with an electrode rather than an injection needle, or can be open at the distal end to allow guide wires or other surgical devices or instruments to be passed through the cannulas.

(26) In the example of treating a patient's bladder with Botox (for OAB) or other medication for other medical conditions, this patent specification highlights the discovery that certain biocompatible polymers can be advantageously configured to inject medication in a desired pattern

of sites over a relatively wide area in the interior wall of a bladder with little or no lateral movement of an introducer. This is made possible in part by using a memory shape, biocompatible polymer cannula with a distal portion that revert to a predetermined curvature after exiting a cannula lumen. One particularly important predetermined curvature has two bends in opposite directions. While it may have seemed counterintuitive that such curvatures of a thin, non-metal cannula would provide enough support for a needle to be pushed into tissue and maintain a desired orientation, this curvature has been proved highly useful and has been particularly helpful in injecting at sites arranged over a relatively large area by only rotating and advancing the cannula, with no or only slight side-to-side motion that can harm tissue. Another significant benefit is that the memory shape polymer used for the cannulas described in this patent specification is much less expensive than the metals such as Nitinol that have been traditionally used as memory shape devices. Other benefits are discussed in said patent and further below.

(27) In the example of using a cannula tipped with an injection needle, the needle can be beveled such that only a distal portion of the bevel is sharpened and cuts a small hole in tissue while a proximal portion of the bevel is less sharp and tends to spread and stretch (dilate), rather than cut tissue around that small hole. This needle shape and structure have the important benefit of allowing tissue at the injection site to contract and close the small hole after the injection needle is withdrawn, to thereby keep the injected medication where it is intended to stay and thereby provide the intended medical benefit and to thereby keep expensive medication from wastefully leaking out of the injection site. In addition, the portion of the injection needle that protrudes distally from the cannula can be shorter than common, and the cannula distal end than surrounds the needle can serve as a penetration depth stop, such that the medication is injected only or to a large extent in the submucosa rather than all the way into the underlying muscle or other tissue. It has been found that the medical benefit of injecting in the submucosa compares favorably with injecting in the underlying muscle or other tissue.

(28) FIG. **1** illustrates a medical injection assembly **1** comprising an introducer **100** that includes a handle **101**, a sheath **102**, and a scope lumen **103** extending from a first proximal end of the handle **101** to a distal end of the sheath **102**, according to some embodiments. The scope lumen **103** is configured to receive an endoscope **110** at the first proximal end of the handle **101** and hold the endoscope in a desired position. A cannula lumen **104** extends from a second proximal end of the handle **101** to the distal end of the sheath **102** and receives a cannula **109** at a second proximal end of the handle **101** and guides the cannula **109** in lumen **104**. The introducer **100** may further comprise a fluid line **105** having a distal end in fluid communication with the scope lumen **103** or with a port at the distal end of introducer **100**.

(29) FIGS. **2** and **3** illustrate scope lumen **103** and cannula lumen **104** in sheath **102** of an introducer **100**, according to some embodiments. Scope lumen **103** may be configured to receive a variety of endoscopes **110** for illuminating and visualizing target tissue within the body. In some embodiments, the endoscope **110** may be a cystoscope. The inside diameter of the scope lumen **103** preferably fits industry standard cystoscopes known in the art. In preferable embodiments, the scope lumen **103** preferably has an inside diameter of about 4 mm to about 5 mm.

(30) The cannula lumen **104** preferably has an inner diameter of about 1 mm to about 2 mm. The inner diameter or lumen **104** is only slightly more than the outside diameter of cannula **109**. The walls of sheath **102** preferably are thin to reduce the outer diameter of the sheath and thereby facilitate passage through a patient's urethra while maintaining strength and rigidity. In preferred embodiments, the sheath walls may have a thickness of about 0.1 mm to about 0.4 mm. The sheath **102** may be comprised of polyether block amides, polyethylene, or other materials with similar rigidity characteristics.

(31) Cannula **109** is a hollow tube made of a biocompatible thermoplastic polymer such as polyether ether-ketone (PEEK), with an outside diameter about 1-2 mm. At least a distal portion of cannula **109** is a shape memory, biocompatible material that reverts to a predetermined curvature

after it projects distally from cannula lumen **104**. The material and the inside and outside diameters of cannula **109** are carefully selected to meet two mutually exclusive requirements” one is to make the distal portion of cannula **109** stiff so it can help push an injection needle into the inner wall of a patient's bladder when said distal portion is curved, for example in an S-shape, and the other is to make that distal portion of cannula **109** flexible so it can be straight when inside a cannula lumen **104**. Preferably, the polymer of cannula **109** has a flexural modulus of about 595,000 psi. Notably, such a flexural modulus allows the user of the device to insert the needle into bladder tissue without causing the cannula to bend or deform in a clinically significant manner. To a skilled person such as a urologist, this means that when injecting in the bladder, the needle would not harm bladder tissue more than has been acceptable in standard-of-care medical practice for conventional injections in the bladder wall and that the visible part of the needle would generally stay in the field of view of a typical endoscope viewing. The bladder wall of adults typically is about 3.5-5.0 mm thick, or an injection needle about 3.0 mm thick would stay in the bladder wall. Some deviation in the needle trajectory during insertion is common and is clinically insignificant so long as it leaves only a temporary tissue disruption, i.e., a disruption that typically clears in a few days, as distinguished from a disruption that punctures the bladder wall or interferes with bladder function for more than a few days. For example, if the cannula deforms so much that the needle cannot penetrate into the bladder wall at a desired injection site or if the cannula bends or deforms so much that the tissue is disrupted to the point of not containing the medication being injected, or if the cannula bends or deforms so much that the needle being forced into tissue opens a slot in tissue rather than a hole about the diameter of the needle, this can be considered a medically significant deformation of the cannula. For example, for a typical adult with a bladder wall thickness about 35-5.0 mm, and a 23-gauge injection needle (about 0.58 mm diameter, typically used to inject medication in the bladder wall) that is 3 mm long from the distal tip of the cannula to the sharp point of the needle (again, typical needle length), a deviation in trajectory within 15-20 degrees from the orientation at the start of insertion would not be clinically significant.

(32) FIG. **3***a* illustrates more clearly an example of a desirable curvature of a distal portion of cannula **109** from which injection needle **114** protrudes distally. This curvature conforms to an inverse tangent function. As illustrated, the distant portion of cannula **109**, which protrudes distally from the distal end of cannula lumen **104** in introducer **102**, has reverted to a curvature that has two bends or inflections in opposite directions and thus can be said to be generally S-shaped. How cannula **109** assumes this curvature is explained below in connection with FIGS. **5**A-C. FIG. **3***b* illustrates an example of an alternative curvature—here the distal portion of cannula **109** that protrudes distally from cannula lumen **104** has reverted to a curvature characterized by a single bend that can be 90 degrees or more and even 180 degrees or more, as may be desirable for a specific medical or other procedure.

(33) FIGS. **3***c-e* illustrate cannula **109** as in may be used alone or with any one of a variety of instruments that can guide it and from which it can emerge distally to revert to a predetermined curvature. Am injection needle **114** can be secured at the cannula's distal tip or the cannula can be used without such a needle. Cannula **109** shown in FIGS. **3***c-e* can be supplied to users alone, with or without an injection needle **114** at its tip. Preferably, cannula **109**, with or without needle **114**, would be supplied in a sterile package that the user can open when needed and use the cannula as desired. As non-limiting examples, a user may pass cannula **109**, with or without needle **114**, through a lumen or working channel of a device such as an introducer or an endoscope or some other guide. Reference numeral **104** schematically illustrates any one of such variety of instruments that may be the same or different from introducer **100** shown in FIG. **1**-**3***b*. As other non-limiting examples, a user may use the cannula of FIGS. **3***c*-**3***e* without an introducer and with or without a needle.

(34) In each of FIGS. **3***c*-**3***e*, cannula **109** is a hollow tube of a shape memory, biocompatible polymer. Several different predetermined curvatures to which a distal portion of the cannula reverts

are illustrated. In FIG. **3***c*, the curvature conforms to an inverse tangent function; in FIG. **3***d*, the curvature is a bend that is over 180 degrees but can be any angle up to an even exceeding 180 degrees; and in FIG. **3***e* the curvature is a single bend.

(35) FIG. **3***e* schematically illustrates a sterile package **105** enclosing cannula **109**; like sterile packages can enclose the cannulas of FIGS. **3***c* and **3***d*. In each case, a needle **114** may be secured to the cannula or may be absent. In some embodiments, the entire instrument illustrated in FIG. **1** may be delivered to a user enclosed in a like sterile pouch, typically without syringe **111**.

(36) The cannula **109** that has been used in a medical procedure typically is discarded as medical grade waste rather than being sterilized and used in another procedure to thereby eliminate or reduce the danger of possible contamination in other procedures.

(37) The predefined curvatures discussed above exist when the cannula is free of external forces that would prevent it from assuming such curvatures. When subjected to external forces, such as when confined in a straight lumen or working channel, the cannula would extend along a straight axis if the lumen or channel is straight or would assume the curvature of the lumen or channel that is not straight.

(38) Returning to the example of using the cannula to inject medication in a patient's bladder, FIG. **4** illustrates handle **101** of an introducer **100** according to some embodiments. A scope lumen **103** extends from a proximal end of the handle **101** and may be configured to receive an endoscope **110** from a proximal end of the handle **101**. A scope seal **112** is positioned at a proximal end of the handle **101** to engage endoscope **110**. The scope seal **112** is a material with an appropriate coefficient of friction to hold endoscope **110** in place. The scope seal **112** preferably comprises silicone.

(39) The cannula lumen **104** extends from a proximal end of the handle **101** and is configured to receive cannula **109** from a proximal end of the handle **101**. A cannula seal **113** is at a proximal end of the handle **101** to engage cannula **109**. The cannula seal **113** preferably is made of a material with an appropriate coefficient of friction to hold a cannula **109** in place. The material of the cannula seal **113** preferably comprises silicone. The handle **101** further comprises a fluid line **105** in fluid communication with the scope lumen **103**. The distal end of the fluid line **105** connects to the scope lumen **103** via a watertight fluid connector. In yet other embodiments, the distal end of the fluid line **105** may be integrated directly into the scope lumen **103** via known manufacturing methods such as various molding techniques, welding, 3D printing, adhesives, etc. The fluid line **105** may comprise a second fluid connector **106**. In preferred embodiments, the second fluid connector **106** may be a Luer lock. The fluid line **105** may further comprise a pinch valve **107**. The pinch valve **107** may control the flow of fluid form a fluid source through the fluid line **105** and into the scope lumen. A port at the distal end of the scope lumen can be provided for outflow or inflow of fluid. FIG. **4** further illustrates a fluid connector **115** coupled to a proximal end of cannula **109** and provided with a pinch valve **113** to selectively deliver fluids into cannula **109** for injecting through needle **114**.

(40) The needle **114** may be a commercially available hypodermic needle suitable for performing injections of OnabotulinumtoxinA or other desirable medication. The outer diameter of the needle **114** is slightly less than the diameter of the cannula **109**, so that the needle can be friction-fitted in the cannula. For example, the needle **114** may be a 23-gauge needle and may extend past the cannula **109** about 1.0 mm to about 3.0 mm in length. In such configurations, the distal tip of the cannula **109** acts as a circumferential wall that helps keep the needle **114** from penetrating into the target tissue past the distal tip of the cannula **109**.

(41) FIGS. **5**A-**5**C illustrate how a cannula **109**, with a needle **114** attached to its distal tip, changes it curvature as it moves distally in cannula lumen **104** such a distal portion of the cannula protrudes distally from the cannula lumen. When the distal portion of cannula **109** is entirely within the cannula lumen, the cannula assumes the shape dictated by the external forces the cannula lumen exerts. If the cannula lumen is straight, so is the distal portion of the cannula while entirely in the

cannula lumen. However, when the cannula **109** is made of the shape memory biocompatible polymer discussed above, as the distal portion of the cannula starts emerging distally from cannula lumen **104**, the part of the cannula that has emerged starts reverting to its predetermined shape. FIG. **5**A illustrates an early stage, in which a relatively short length of cannula **109** has emerged and has reverted to a curvature of a single bend down. FIG. **5**B illustrates a stage in which a greater length of cannula **109** has emerged and still has only a single bend. FIG. **5**C illustrates a stage in which the entire desired distal portion of cannula **109** is out of cannula lumen **104** and has reverted to a predetermined shape conforming to an inverse tangent function, with two bends of inflections in opposite directions.

(42) The vertical distance between the distal tip of cannula **109** and a long axis of cannula lumen **104** changes in the course of movement of cannula **109** distally out of cannula lumen **104**. This distance is relatively small in FIG. **5**A, greater in FIG. **5**B, and greatest in FIG. **5**C. One benefit of this feature is discussed in connection with FIGS. **8**A-B.

(43) FIGS. **6-7** illustrate a syringe **111** that can be coupled to fluid connector **115** for delivery of fluid such as medication for injection in a patient's bladder through cannula **109** and needle **114**. Syringe **111** comprises a syringe barrel (not seen in these FIGs.) and a plunger body **116** having a first portion **117** proximate the proximal end and second portion **118** proximate the distal end, wherein the first portion **117** has a plurality of corresponding detents **119** on opposite sides of the first portion **117**. The syringe **111** may also comprise a sealing cap **120** attached to the distal end of the plunger body **116** and a finger grip **121** comprising two paddles **122**. The finger grip may be configured to be removably coupled to the plunger body **116** and to interact with the detents **119** to provide audible and tactile feedback to a user when the plunger body **116** is pushed through the finger grip **121** in a distal direction. The second portion **118** may fit into commercially available syringe barrels with the sealing cap **120** forming a watertight seal within the syringe barrel. In preferred embodiments, commercially available 10 cc syringe barrels may be used. The finger grip **121** may be configured to clip on to the plunger body **116**. The finger grip **121** may include tabs **123** on the interior of both paddles **122**. The tabs **123** may fit into grooves between the detents **119** on the first portion **117**. When the user pushes the plunger body **116** in a distal direction into the syringe barrel, the detents **119** provide resistance against the movement until the tabs **123** bend enough to clear a set of detents **119** and fit into the next set of grooves. The detents **119** may be spaced along the first portion **117** such that clearing one set of detents **119** results in an ejection of a specific amount of fluid from the syringe. In preferred embodiments, clearing one set of detents would result in the ejection of 1 cc of fluid from the syringe. When the user causes tabs **123** to clear a set of detents **119** and the tabs **123** come to rest in the subsequent grooves, the user is provided with tactile and audible feedback to indicate that one such predetermined unit of fluid has been ejected from the syringe.

(44) FIGS. **7***a* and **7***b* are two perspective views from different viewpoints that illustrate another example of a syringe that can be coupled to fluid connector **115**. A syringe barrel **300** has a coupler **305** at a distal end that is shaped and dimensioned to fit connector **115** and a finger hold **301** at a proximal end that has a paddles or wings **306**, **307** and a flexible tab **302**. A plunger body **303** slides in barrel **300** and has at its distal end a cap **308** that forms a seal against the inside wall of barrel **300**. Plunger body **303** has detents **304** along at least one of its long walls that flexible tab **302** engages as the plunger body moves relative to the barrel.

(45) FIGS. **8**A-**8**B illustrate therapeutically effective patterns of injection sites according to some embodiments. It has been found beneficial to disperse the injections of medication such as OnabotulinumtoxinA across the bladder tissue. In preferred embodiments, injections patterns may comprise three concentric semi-circles in the lower half of the bladder with radii A, B, and C. Such an injection pattern may be created by moving the cannula **109** distally until the needle **114** is at a distance A from the long axis of the sheath **102** of the introducer **100** and rotating one or both of introducer **100** and cannula **109** until the needle **114** is at injection site **200**. The introducer **100**

and/or cannula **109** are then moved distally to penetrate the bladder's inner wall with injection needle **114** and inject OnabotulinumtoxinA or other medication. The needle **114** is then withdrawn from the bladder wall by proximally moving cannula **109** alone or together with introducer **100** and the introducer and/or the cannula are rotated about the long axis of the introducer until the needle **114** is at injection site **201**, and the injection process is repeated. Once the injection pattern for the semi-circle with radius A is complete, the cannula **109** may be moved distally relative to introducer **100** (and the introducer may be moved proximally or distally as needed) until the needle **114** is at a distance B from the axis defined by the sheath **102** of the introducer **100**, and the previous steps can be repeated to create the injection patterns for the semi-circles with radii B and C. Preferably, A is approximately 0.43 inches, B is approximately 0.8 inches, and C is approximately 1.2 inches. By rotating the introducer **100** and/or cannula **109** about the introducer's long axis to position the needle rather than moving the introducer **100** laterally or pivoting it about an axis transverse to the long axis, the patient experiences less discomfort and possible injury from stretching of the urethra.

(46) A method for treating overactive bladder comprises inserting an endoscope **110** into a scope lumen **103** of an introducer **100**, according to some embodiments. The method may further comprise inserting a cannula **109** into a cannula lumen **104** of the introducer **100**, the cannula **109** configured such that the distance between the distal tip of the cannula **109** and the axis defined by the sheath **102** of the introducer **100** increases as the cannula **109** is moved in a distal direction out of the introducer, wherein a syringe **111** filled with OnabotulinumtoxinA or other medication is coupled to the proximal end of the cannula **109**. The method may further comprise guiding the introducer **100** through the urethra of a patient to the patient's bladder. The method may further include extending the distal portion of the cannula **109** past the distal end of the introducer **100** until a needle **114** attached to the distal end of the cannula **109** is placed at a desired radial distance from the long axis defined by the sheath of the introducer. The method may further comprise rotating the introducer **100** and/or cannula **119** to position the needle **114** at a desired position. The method may further comprise moving the introducer and/or the cannula in a distal direction to insert the needle **114** into the bladder. The method may further include activating the syringe **111** to inject OnabotulinumtoxinA or other medication into the bladder. The method may further include moving the introducer **100** and/or the cannula **109** in a proximal direction to withdraw the needle **114** from the bladder. The method may further include repeating the extending, rotating, moving distally, activating, and moving proximally steps until a therapeutically effective amount of OnabotulinumtoxinA or other medication has been injected in a therapeutically effective pattern into the bladder.

(47) An alternative method comprises enclosing a cannula **109**, with or without an injection needle **114**, in a sterile pouch and delivering the pouch to a user. The method further comprised opening the sterile pouch and passing the cannula through an lumen or channel of a device such as an introducer, endoscope or some other guide that may or may not be already inserted into a patient's cavity. The method further comprises advancing the cannula distally until a distal portion thereof starts protruding distally from such lumen or channel and begins to revert to its predetermined curvature. The method further includes advancing the cannula distally even more relative to a distal end of the lumen or channel, so more of the distal portion of the cannula reverts to its predetermined shape and using the cannula such as to inject a substance into a patient's bladder or deliver or inject a substance into another body cavity. The cannula is them withdrawn from the patient, alone or together with the instrument through which it passed, and the cannula **109** is discarded along with any needle **114** as medical grade waste.

(48) FIG. **9** illustrates in a perspective view an injection needle **90** protruding from a cannula **94** that has a distal end **94***a*. Injection needle **90** has a distal portion **90***a* that is sharpened to easily penetrate tissue to make a small hole, much smaller than the diameter of needle **90**. A proximal portion **90***b* is less sharp and can be deburred such that it would dilate rather than cut tissue at the small hole that the sharpened distal portion **90***a* has made. For example, the portion of needle **90**

that protrudes from cannula **94** can be roughly 3 mm long, and only roughly 0.5 mm at the distal end can be sharpened while the rest of the beveled portion is sufficiently dull so it would dilate rather than cut tissue. The entire beveled portion can be roughly 2.1 mm from the distal tip to the proximal end of the beveled portion, in the case of a 23-gauge injection needle. The bevel angle can be roughly 17 degrees. For different-gauge injection needles or for different medical procedures or needs, these dimensions may vary.
(49) In use to inject medication in tissue such as a bladder wall, it has been observed that injecting medication such as Botox with a needle such as **90**, which makes a small hole with its tip and dilates tissue as the remainder of the bevel enters tissue, results in a well-defined bleb (a fluid-containing, elevated lesion or blister) that does not leak out the injected medication after the needle is withdrawn, or leaks significantly less than a case of an injection that is otherwise the same but uses a needle with a bevel that cuts rather than dilates tissue.
(50) FIG. **10** illustrates in a sectional view an example of an injection needle, with indicated examples of dimensions.
(51) FIG. **11** is otherwise the same as FIG. **3**A but shows a device or instrument **114***a* protruding from cannula **109**. Device or instrument **114** a can be an electrode or an ablation needle, for example a Bugbee or Bovie type electrode, or a guide wire that can move within cannula **109**, or another surgical device or instrument. A cannula **109** or **94**, with an injection needle **114** or **90** at its distal end, or with a device or instrument **114***a* at its distal end, or with an open distal end without any device in the cannula, can serve in a variety of medical fields and procedures, including cystoscopy, hysteroscopy, ureteroscopy, colonoscopy, electrosurgery, endometrial ablation, myeomectomy, intratubal access, tubal sterilization, biopsy (uterine, bladder, etc.), implanting medication, and OAB Botox injections, to name a few. The electrode or device **114***a* can apply electromagnetic energy such as RF energy to tissue, or ultrasonic energy, or heat.
(52) In the example of using a cannula such as **94** or **109** with an injection needle such as **90** or **114**, the curve at the distal portion of the cannula provides a well-controlled injection angle such that the injection needle can be perpendicular or close to perpendicular to the wall of the tissue at the injection site or another device such as an ablation electrode also can be oriented at a preferred angle. The curve at the distal portion of the cannula has a further benefit—it allows a viewing device such as an endoscope through which the cannula passes to observe a greater viewing field of motion of the needle compared with conventional injection needles, and to require no or minimal motion of the endoscope or introducer relative to the patient, thus providing greater patient comfort because of less torquing or other motion relative to the patient.
(53) The distal portion of the cannula, with two bends in opposite directions, preferably is sufficiently stiff for the intended medical procedure. For example, if the cannula is used to inject medication in tissue, the stiffness should be enough to make the injection needle penetrate the type of tissue at the injection site when pressed against the tissue by action on a portion of the cannula that is proximal to the distal portion with the two bends. The selected stiffness may differ for different types of tissue at the intended injection sites. In another example, where an ablation electric stylet is used rather than an injection needle, the distal portion of the cannula (with the two bends) may be less stiff as the medical procedure may require less pressure of the stylus on tissue. When yet different devices protrude from the distal end of the cannula, different stiffness parameters may be selected for the cannula's distal portion, appropriate for the intended medical procedure.
(54) Although the foregoing has been described in some detail for purposes of clarity, it will be apparent that certain changes and modifications may be made without departing from the principles thereof. There can be many alternative ways of implementing both the processes and apparatuses described herein. Accordingly, the present embodiments are to be considered as illustrative and not restrictive, and the body of work described herein is not to be limited to the details given herein, which may be modified within the scope and equivalents of the appended claims. Numerous

specific details are described to provide a thorough understanding of the disclosure. However, in certain instances, well-known or conventional details are omitted to avoid obscuring the description.

Claims

1. A medical device for insertion into a patient's body, said device comprising: an elongated hollow cannula made of a biocompatible, non-metallic material; said cannula having a first shape when confined by first external forces acting thereon; said cannula having a distal portion that reverts to a second shape, different from the first shape, when free of said first external forces, and said distal portion having a distal end; said second shape having a predetermined curvature defined by an inverse tangent function and comprising two bends in different directions; whereby said distal end of the cannula is radially spaced from an axis of said first shape when said distal portion of the cannula has reverted to said second shape; and wherein said cannula is sufficiently flexible to assume said first shape when in a working channel of an introducer or endoscope but at least said distal portion of the cannula has a flexural modulus that is sufficiently high when in said second shape to force said distal end against tissue defining an internal wall of an internal body cavity for selected medical action on said tissue when a selected distal force is exerted on a proximal portion of the cannula by the introducer or endoscope.
2. The device of claim 1, further including an injection needle protruding from said distal end of the cannula, wherein said distal portion of the cannula has a sufficiently high flexural modulus for inserting said needle into said tissue after the distal portion of the cannula has reverted to said second shape.
3. The device of claim 1, further comprising a surgical device protruding distally from said cannula.
4. The device of claim 1, in which at least said distal portion of said cannula has a flexural modulus of about 595,000 psi.
5. The device of claim 1 in which an entirety of the cannula is a single hollow tube made of PEEK.
6. The device of claim 1, in which said two bends of the distal portion of said cannula are spaced from each other by a straight portion of the cannula.
7. The device of claim 1, in which said two bends are in opposite directions.
8. A method comprising: inserting an elongated hollow cannula made of a biocompatible, non-metallic material into a passage in a patient's body while confining at least a distal portion of said cannula to a first shape by first external forces acting thereon; thereafter gradually moving a distal end of said cannula distally from said confining to thereby: cause said distal portion of said cannula to gradually revert to a second shape, different from the first shape, that has a predetermined curvature defined by an inverse tangent function and comprises two bends in different directions; and cause said distal end to move radially from an axis of said first shape; and applying medical action at selected sites of tissue defining an internal wall of an internal body cavity of said patient with a device protruding distally from said distal end of said cannula when said cannula is no longer confined in said first shape; wherein said selected sites are spaced from each other by distances that depend on a distal distance of said device from said first external forces.
9. The method of claim 8, wherein said confining comprises enclosing said cannula in a working channel of an introducer or endoscope.
10. The method of claim 8, in which said medical action comprises injecting medication at said sites.